

PROJECT IDEA SUBMISSION

PROJECT TITLE:

**PhishShield: A Machine Learning-
Based Web Application for
Phishing URL Detection**

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ABSTRACT:

Phishing attacks are among the most common cyber threats, where attackers create fake websites or URLs to steal sensitive information such as passwords, banking credentials, and personal data. Existing detection methods often rely on blacklists, which cannot identify newly generated phishing websites.

This project proposes a web application that detects phishing URLs using machine learning techniques by analyzing various URL-based features. Users can enter a website URL into the application, and the system predicts whether it is legitimate or phishing in real time.

The application is designed with a user-friendly interface and provides fast and reliable results to help users avoid online fraud. By integrating machine learning with a web-based platform, the proposed system aims to improve cybersecurity awareness and reduce the risk of phishing attacks.

Introduction:

With the rapid growth of online banking, e-commerce, and digital communication, phishing attacks have become one of the most dangerous cyber threats. Cybercriminals create fake websites that closely resemble legitimate ones to deceive users into revealing confidential information. Since phishing websites can be created quickly, traditional blacklist-based methods often fail to detect newly generated malicious URLs.

Machine learning provides an intelligent approach by analyzing the characteristics of a URL and identifying suspicious patterns. This project develops a web application that allows users to verify the authenticity of URLs before accessing them, thereby improving online safety.

Problem Statement:

Internet users frequently encounter fraudulent websites designed to steal sensitive information. Many phishing websites remain undetected because they are newly created and are not yet included in blacklist databases. Existing security solutions may also require browser extensions or paid software

.

Therefore, there is a need for a lightweight, accessible, and intelligent web application capable of detecting phishing URLs accurately in real time.

Objectives:

- **Develop a web application for phishing URL detection.**
- **Extract meaningful features from URLs.**
- **Train a machine learning model using phishing and legitimate URL datasets.**
- **Predict whether a URL is safe or malicious.**
- **Provide a simple and user-friendly interface.**
- **Improve cybersecurity awareness among users.**

Proposed Solution:

The proposed system is a web application that predicts whether a given URL is phishing or legitimate using a trained machine learning model.

The system workflow includes:

- User enters a URL.
- URL features are extracted.
- The trained model analyzes the features.
- The prediction is displayed.
- The user receives a security recommendation.

EXPECTED OUTCOMES:

- **Accurate phishing URL detection.**
- **Easy-to-use web interface.**
- **Faster identification of malicious websites.**
- **Increased cybersecurity awareness.**
- **Reduced phishing attacks on users.**

FUTURE SCOPES:

- **Browser extension integration.**
- **Mobile application support.**
- **Real-time website content analysis.**
- **Deep learning-based detection.**
- **Continuous model updates using new phishing datasets.**
- **Multilingual phishing detection.**

EXPECTED BENIFITS:

- **Improved cybersecurity.**
- **Easy access through any web browser.**
- **Cost-effective solution.**
- **Real-time phishing detection.**
- **Educational tool for cybersecurity awareness.**

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